

Metric	ScholarQABench			SQuALITY			LexAbSumm		
	$w = 0$	$w = 1$	$w = 2$	$w = 0$	$w = 1$	$w = 2$	$w = 0$	$w = 1$	$w = 2$
BARTScore	0.03	0.03	0.02	0.03	0.03	0.03	0.15	0.16	0.16
MiniCheck	0.17	0.15	0.15	0.11	0.15	0.19	0.47	0.53	0.60
SummaC-Conv	0.22	0.23	0.25	0.22	0.23	0.24	0.33	0.33	0.34
SummaC-ZS	0.14	0.18	0.20	0.11	0.13	0.14	0.36	0.38	0.39
AlignScore	0.15	0.21	0.27	0.10	0.18	0.24	0.36	0.52	0.64
UniEval	0.72	0.73	0.74	0.67	0.68	0.70	0.81	0.82	0.84

Table 1: Impact of retrieval context window size w on factuality scores for original summaries. Each value reports the average factuality score assigned by a given metric over the dataset, computed using a retrieval-based setup where each summary sentence is evaluated against a retrieved source sentence and its surrounding context of size w . Increasing w expands the number of neighboring sentences included around each retrieved sentence, providing a broader local context for factuality assessment.

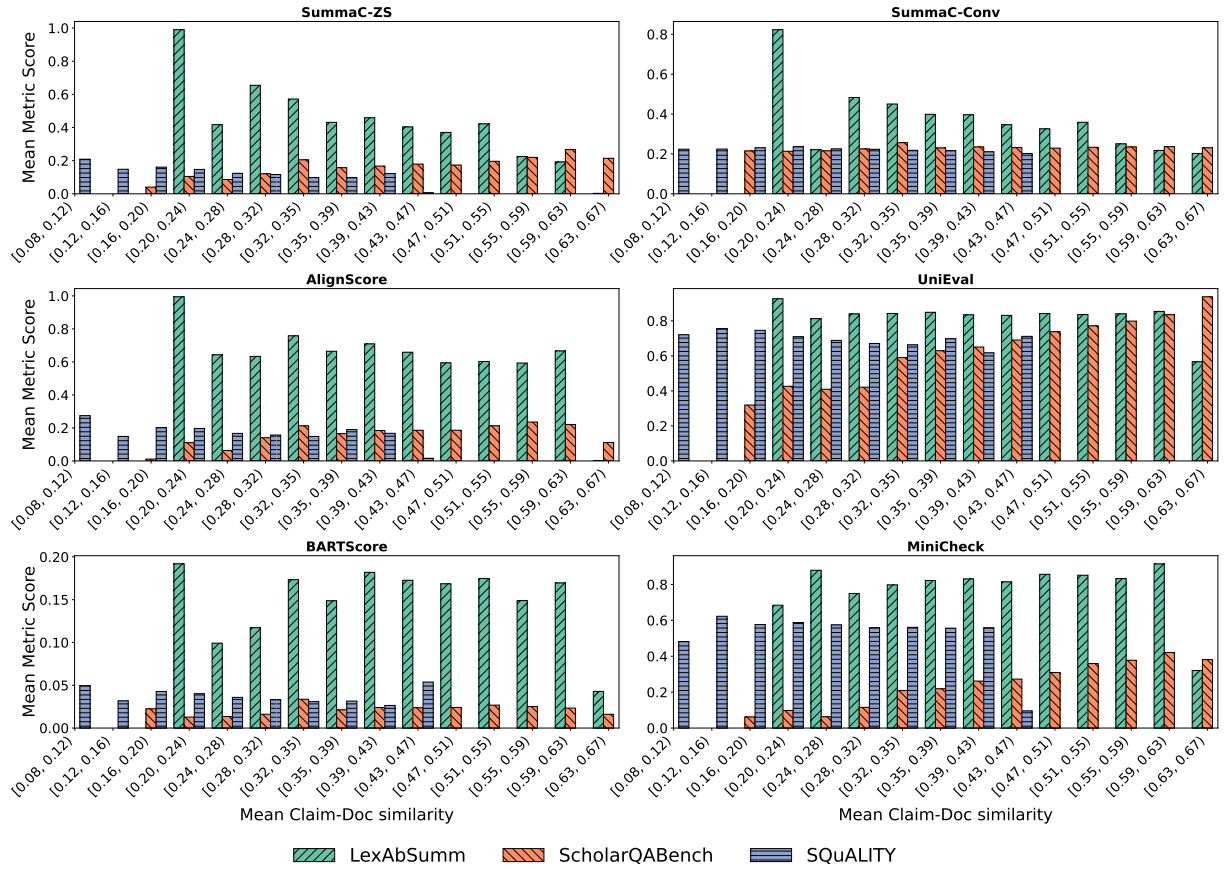


Figure 3: Relationship between claim similarity and average factuality score. Higher similarity values correspond to more information-dense claims whose content overlaps with multiple parts of the source document. Metrics generally assign lower scores to these claims for LexAbSumm and SQuALITY, and higher scores for ScholarQABench, indicating reduced reliability for compressed information.

ric reliability. We hypothesize that claims which are highly similar to many claims in the original document may reflect a form of *hubness* in high-dimensional embedding spaces (Samir et al., 2024; Radovanovic et al., 2010; Lazaridou et al., 2015), where a sentence appears broadly similar to many others without being cleanly grounded in any single

supporting span. Such claims tend to contain more general (and thus compressed) information, and so are harder to fact-check since evidence for them is dispersed across multiple parts of the source document (Goldman et al., 2024).

We approximate this by computing the mean pairwise cosine similarity between each summary

sentence s_j (claim) and all n sentences d_i in the source document D ,

$$\text{Sim}(s_j, D) = \frac{1}{n} \sum_{i=1}^n \cos(e_j, e_i^D), \quad (3)$$

where e_j and e_i^D denote the sentence embeddings of the claim and document sentences, respectively. Higher $\text{Sim}(s_j, D)$ values indicate more broad claims whose meaning overlaps widely with the source, while lower values correspond to specific, localized claims. Claims are then grouped into similarity bins $\mathcal{B}$, and the average factuality score for each bin is calculated as

$$\text{Score}_{\text{bin}} = \frac{1}{|\mathcal{B}|} \sum_{s_j \in \mathcal{B}} M(s_j), \quad (4)$$

where $M(s_j)$ is the factuality score for claim s_j .

A few trends emerge between claim similarity and metric sensitivity across all datasets and metrics, as shown in Fig. 3. For *LexAbSumm*, metric scores consistently decrease as claim similarity increases, meaning that the more a claim’s meaning is entangled with the broader document, the worse the metric becomes at predicting factuality. This is likely because summaries of legal documents may refer to specific aspects of the texts, which are easier to fact-check, while more general statements which compress a lot of technical language are more difficult. The same occurs (though less pronounced) for *SQuALITY*, where more general claims are more challenging to fact check. In this case, we are summarizing novels, so more general claims try to compress the story narrative into a compact form, and thus will require retrieving or attending to and reasoning over disparate pieces of the text.

This effect is particularly visible in *AlignScore* and *BARTScore*, both of which depend on local lexical alignment or sentence-level contextual matching. Their scores show sharp declines for high-similarity claims, reflecting their sensitivity to distributed evidence. *SummaC-Conv* and *SummaC-ZS*, while somewhat more stable, also exhibit a gradual drop as similarity increases, which suggests that NLI-based judgments still rely on relatively localized entailment cues. In contrast, *UniEval* and *MiniCheck* maintain comparatively stable performance across bins, implying a higher degree of robustness to distributed or compressed content, although some degradation is still observed in the highest-similarity regions.

On the contrary, we see that more general statements are easier to fact-check in *ScholarQABench* for many metrics. This could be because *ScholarQABench* is multi-document, so many sentences being similar to one claim may actually simply be repeated instances of the claim across documents. We see this upward trend especially on *UniEval* and *MiniCheck*, where metric quality is highly dependent on how general or specific a claim is.

On the whole, these findings support the broader conclusion that factuality metrics are dependent on how dispersed and overlapping the evidence is for a given claim, a hallmark of long-document summarization. Improving robustness for such cases likely requires metrics that can reason over multi-span evidence rather than relying on local or pairwise semantic alignment.

6 Conclusion & Future Work

In this work, we present a comprehensive evaluation of six widely used reference-free factuality metrics: *BARTScore*, *SummaC-Conv*, *SummaC-ZS*, *AlignScore*, *MiniCheck*, and *UniEval*. We tested their behavior under seven meaning-preserving perturbations applied to long-document summaries to assess whether these metrics reliably capture factual consistency. To enable evaluation over long documents, we used a sentence-level retrieval-based scoring strategy, which compares each summary sentence to the most relevant evidence snippets from the source document. This setup enabled fine-grained evaluation across three diverse long-form abstractive summarization datasets: *SQuALITY*, *LexAbSumm*, and *ScholarQABench*, covering sci-fi, legal, and scientific domains.

Our results revealed that many metrics respond inconsistently to perturbations that do not affect factual consistency. Several metrics exhibit unstable behavior in response to paraphrasing, simplification, and logically equivalent negations. *AlignScore* and *SummaC-ZS* are particularly unreliable across domains and perturbation types. In contrast, *UniEval* and *MiniCheck* are relatively robust, although they too struggle in specific cases, such as handling logical negations. Most metrics improve when evidence retrieval windows are expanded, particularly for complex, multi-sentence inputs such as legal documents. We also found that metrics are systematically affected by the information-density of claims whose meaning overlaps broadly with the source document, indicating that current ap-

proaches struggle to evaluate compressed or contextually entangled statements, which are common in long-form summarization. These findings highlight a need for factuality metrics that are robust to stylistic and logical variation, retrieval-aware, and sensitive to information density.

Future work should explore multi-span reasoning and context-aware calibration to better model distributed evidence, as well as contrastive training on meaning-preserving perturbations to improve stability. Incorporating human judgments can identify systematic weaknesses and support the design of metrics that generalize across domains and languages. We also see potential in hybrid approaches that combine reference-free with reference-based alignment signals, bridging semantic precision with contextual coverage for more reliable evaluation of long-document summarization. Additionally, extending perturbation strategies to better capture long-document phenomena, such as evidence relocation and cross-reference disruption, offers a promising direction for stress-testing long-range coherence and evidence tracking.

Limitations

While our study offers a systematic investigation into the robustness of factuality metrics under meaning-preserving perturbations in long-document summarization, there are several aspects that merit further consideration. Our analysis relies on automatically generated perturbations, produced using GPT-4o, which are designed to preserve factual consistency. However, without human annotations, we cannot confirm with full certainty that all edits preserve factual correctness in every case. We also do not evaluate metric outputs against human factuality judgments in the long-document setting. Large-scale human annotations for long-document summaries are currently scarce, and conducting such an evaluation would require the creation of a new benchmark with human judgments across multiple metrics, domains, and perturbation types. This represents a substantial research effort beyond the scope of this work. This may introduce noise into the interpretation of metric behavior, especially when changes are subtle or domain-specific. We evaluate six reference-free metrics in their original, publicly released form and do not investigate whether fine-tuning, calibration, or adaptation to long-form inputs might mitigate some of the observed weaknesses. In our retrieval-based scoring

setup, we use a fixed number of top-k most similar sentences and vary only the surrounding context window to control retrieval granularity. While this gives us insight into how context affects metric behavior, it assumes a static retrieval strategy and does not account for dynamic query-based retrieval or more sophisticated evidence selection methods that may better match human annotation patterns. Additionally, our analysis is confined to English-language datasets from three domains: science fiction, legal text, and scientific articles. These domains offer diversity in structure and style, but our findings may not fully generalize to other high-stakes applications such as medical or financial summarization, or to non-English and low-resource settings. Addressing these broader limitations will be important for future work aiming to build more generalizable and reliable factuality evaluation pipelines.

Ethical Implications

This study evaluates automatic factuality metrics rather than developing new summarization models, and thus presents minimal direct ethical risk. However, factuality evaluation plays an important role in ensuring the reliability of language model outputs. Weak or biased metrics could inadvertently overestimate the truthfulness of generated content, particularly in sensitive domains such as medicine or law. By identifying systematic weaknesses and proposing strategies for more reliable evaluation, this work aims to support safer and more accountable deployment of summarization systems. All datasets used in this study are publicly available and contain no personally identifiable information.

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References

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